BOOK

LINUX LAB

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1 Introduction to Computer Systems and Operating Systems

1 Computer System Structure (Hardware, OS, Apps, Users)

📖 DEFINITION 1: **Computer System**

A computer system is a set of hardware devices and software that work together to execute and manage various operations and tasks within computers. A computer system consists of a variety of components that cooperate to provide an execution environment for applications and perform computing operations.

A computer system can be divided into four main components:

- **Hardware:** The physical components of the computer, such as the Processor, Memory, Storage Devices, I/O Devices, and so on.
- **Operating System:** A collection of software, each designed for a specific function, aimed at controlling and coordinating the use of hardware among various applications and users.
- **Application Programs:** Software that defines the ways in which system resources are used to solve computing problems for users (e.g., Word Processors, Database Systems, Web Browsers, Compilers).
- **Users:** The entities using the computer, which may be people, machines, etc.

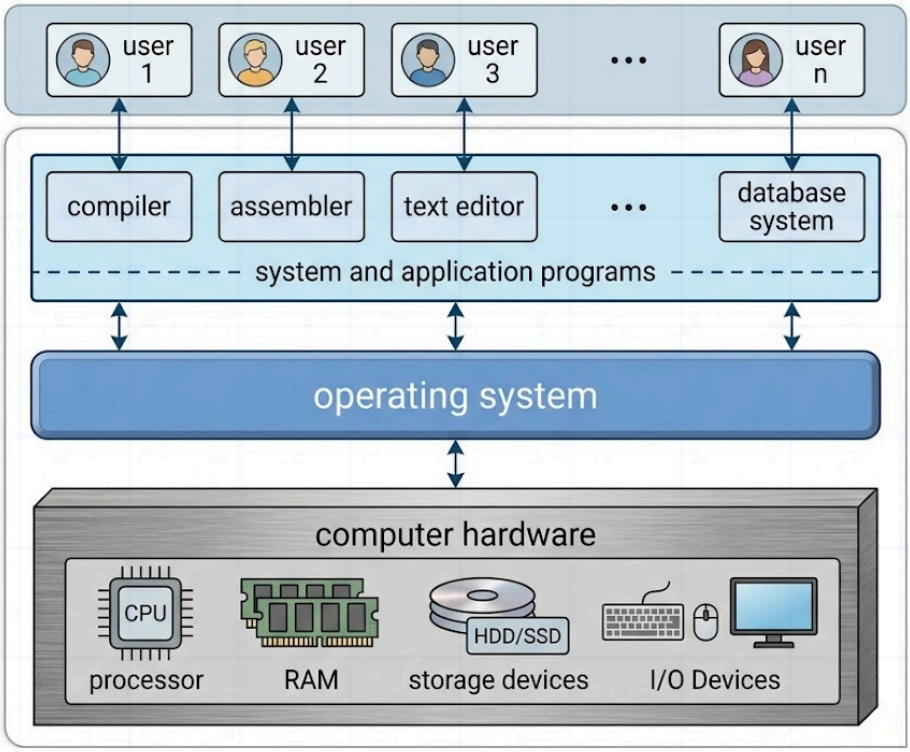


Figure 1: Computer System Components

2 Introduction to Operating Systems

2.1 What is an OS? (Intermediary layer)

An operating system is necessary in every computer to run programs and manage resources. It organizes and coordinates the operation of various hardware components in the computer, such as the processor, memory, storage units, and input/output units.

📖 DEFINITION 2: **Operating System (OS)**

An **Operating System (OS)** acts as an intermediary layer between computer hardware and the software applications executed by users. Its main objective is to provide a safe, efficient, and user-friendly environment where applications can run without needing direct control over low-level hardware routines.

Table 1: *Primary Subsystems and Responsibilities*

Subsystem	Primary Function	Core Abstractions / Concepts
Process Management	Allocates CPU time to competing threads and handles process state transitions.	PCB (Process Control Block), Context Switching, Preemptive Scheduling
Memory Management	Maps virtual addresses to physical RAM and manages secondary memory paging.	Virtual Memory, Page Tables, MMU (Memory Management Unit), Swapping
Storage & File Systems	Provides structured file abstraction and manages non-volatile disk blocks.	Inodes, VFS (Virtual File System), Allocation Tables (FAT, NTFS, ext4)
I/O & Device Drivers	Buffers, caches, and routes requests between hardware devices and applications.	Interrupt Handlers, DMA (Direct Memory Access), Character/Block Devices
Security & Access Control	Enforces user privileges, resource isolation, and access permission rules.	Access Control Lists (ACLs), User vs. Kernel Mode (Ring 0 / Ring 3)

2.2 OS Services (User-facing vs. System-facing)

To achieve this, the OS architecture is structured around two key categories of services:

- **User-Facing Services:** Capabilities designed to help users and application programs carry out tasks effectively (such as process execution, input/output management, and file storage).
- **System-Facing (Efficiency) Services:** Background functions designed to ensure the underlying system operates smoothly, fairly, and securely (such as allocating resources among competing processes, error handling, and security enforcement).



Figure 2: Typical Operating System architecture diagram

1. User Interface (UI) Layer

The top layer of the Operating System allows users and application programs to interact with the system through three main interfaces:

- **Graphical User Interface (GUI):** A visual interface using windows, icons, and menus for interactive input.
- **Command Line Interface (CLI):** A text-based interface where users type direct commands for process or file execution.
- **Batch Interface:** An automated execution mechanism where series of commands/programs are queued and executed without direct user intervention.

2. User-Facing Services

These services directly assist users and application programs in executing tasks:

- **Program Execution:** Loads programs into memory, handles execution, manages processes, and terminates execution normally or handles abnormal crashes.
- **I/O Operations:** Provides a standardized interface for running programs to communicate with input/output hardware (keyboards, disks, displays, printers) without managing hardware-level device drivers directly.
- **File-System Manipulation:** Handles creating, reading, writing, searching, deleting, and updating files and directories, along with setting access permissions.
- **Communications:** Enables processes to exchange information either on the local machine (inter-process communication via shared memory or message passing) or across network links.

3. System-Facing (Efficiency & Management) Services

These background services ensure the operating system operates efficiently, fairly, and securely among multiple users or processes:

- **Resource Allocation:** Manages and allocates system hardware resources—such as CPU execution cycles, main memory (RAM), and storage devices—among competing concurrent jobs.
- **Accounting:** Tracks and logs resource utilization metrics (CPU usage time, memory overhead, disk usage) for system monitoring, optimization, or billing.
- **Error Detection:** Continuously detects and handles errors occurring in hardware (memory errors, power failures), I/O devices, or software bugs to maintain system stability.
- **Protection and Security:**
  - **Protection:** Ensures internal process isolation so concurrent processes cannot corrupt each other's memory space or system resources.
  - **Security:** Controls access to system resources via user authentication, credentials, and access control permissions to prevent unauthorized external access.

> **Key Distinction**
>
> The fundamental core program that runs at all times on the computer is the **Kernel**. Everything else is classified as either a system program or an application program.

2.3 System Calls

During program execution, an application may require services provided by the operating system (e.g., accessing hardware, reading/writing files, or managing processes). To access these services, the OS provides **System Calls**. These system calls are executed and handled within **Kernel Mode**.

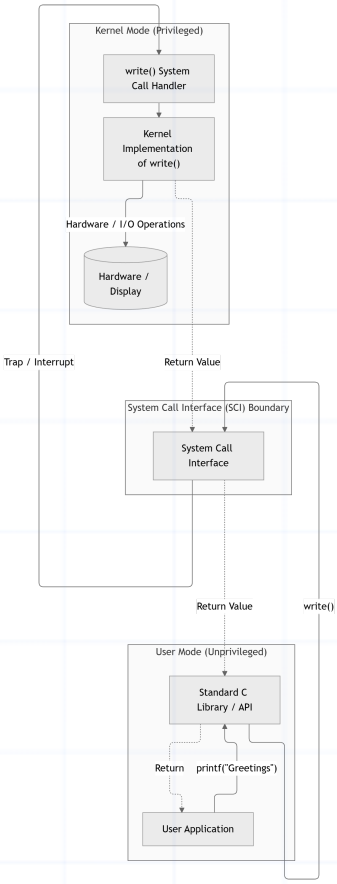


Diagram 1: System Call Architecture Diagram

2.3.1 How System Calls Work

When a program requests a service like printing, specific underlying functions (system calls) are invoked in the background. Parameters are passed to these functions, allowing the operating system to safely execute the requested operation.

- **Mode Execution:** System calls operate in **Kernel Mode** to ensure security and prevent user programs from directly manipulating critical hardware or memory.
- **Execution Frequency:** Modern operating systems execute thousands of system calls per second.
- **Common Linux System Calls:** `open`, `read`, `write`, `close`, `wait`, `exec`, `fork`, `exit`.

2.3.2 APIs vs. System Calls

Directly writing code with raw system calls can be complex and platform-dependent. Therefore, application developers typically use an **Application Programming Interface (API)** operating in **User Mode** (such as POSIX API in C or the Standard C Library).

Behind the scenes, API functions invoke the corresponding system calls via the **System Call Interface (SCI)**.

- **Example (Printing to Screen):**
  1. The user application calls `printf("Greetings");` via standard C library in **User Mode**.
  2. The function triggers `write()` through the **System Call Interface (SCI)**.
  3. The system switches to **Kernel Mode** to execute the system call and complete the output operation.
  4. The result and return status are passed back across the boundary to **User Mode**.

2.4 Operating System Operations

Operating systems execute a variety of operations across both hardware and software levels to maintain system stability, performance, and security.

Key operations include:

- **Interrupt Handling (Interrupts):** Operating systems use interrupts to manage asynchronous and unexpected events at the hardware level. When an interrupt occurs—such as a hardware request or software error—the OS pauses the current execution flow to handle the event immediately. This mechanism ensures quick responsiveness to critical system events and enables efficient resource scheduling.

- **Software Error Management:** Operating systems detect and gracefully manage errors or exceptions caused by running applications (such as a division-by-zero error). Instead of allowing the system to crash, the OS intercepts the error, provides an appropriate status or error message, and safely terminates or handles the faulty process.
- **System Service Requests:** Applications request specific services from the operating system to perform low-level or restricted tasks. These include file operations (creation/deletion), memory management, network communication, process management, and access to hardware devices like storage drives or printers. The OS processes these requests securely and efficiently.

2.5 Operating System Modes (OS Modes)

Operating systems implement dual-mode operation (**Dual Mode**) to separate and protect critical system tasks from user applications, ensuring system stability, protection, and security.

- **User Mode:** When user applications execute—such as editing a document or running software—the CPU operates in **User Mode**. In this mode, application execution privileges are restricted and isolated from core system resources and hardware routines. This isolation prevents user programs from inadvertently crashing the system or directly interfering with vital OS processes.
- **Kernel Mode (Supervisor Mode):** When an application requests a service from the operating system (via a **system call**) or when a hardware interrupt occurs, the CPU switches execution mode from User Mode to **Kernel Mode**. In Kernel Mode, the operating system has unrestricted, full access to all underlying hardware components, physical memory, and privileged system instructions. This mode handles sensitive operations such as memory management, CPU scheduling, and hardware device communication.

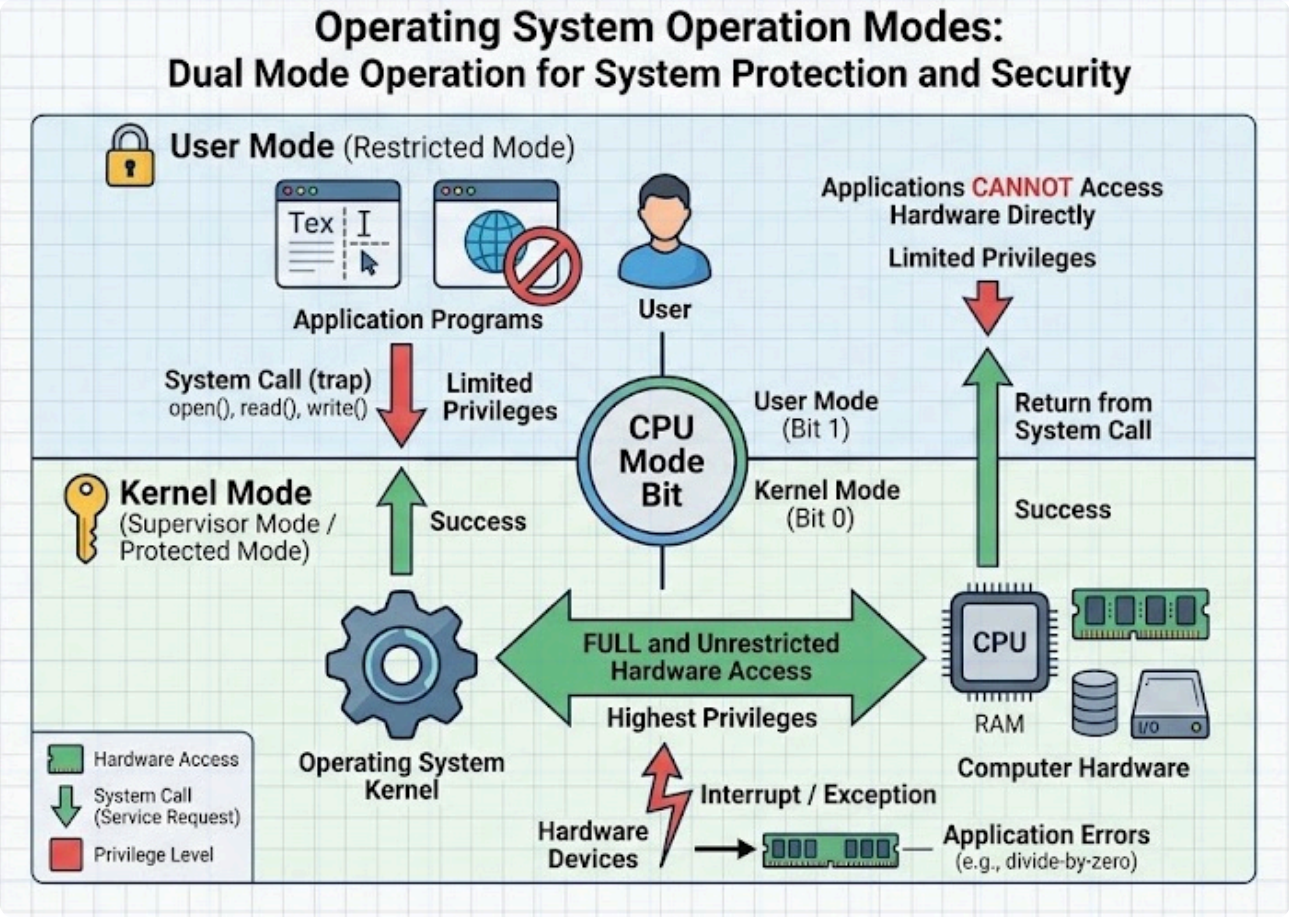


Figure 3: The key distinctions between User Mode and Kernel Mode, showing how privileges are handled

Key Takeaways

- ⭐ The **Dual Mode** mechanism provides fundamental security and isolation.
- ⭐ Restricted operations are strictly contained in User Mode, while privileged and hardware-level operations are safely executed within Kernel Mode.

3 Command Line Interface (CLI)

The Command Line Interface (CLI)—also known as the command interpreter—is an interface that allows users to interact with the operating system and software programs using text-based commands. The CLI works by parsing, analyzing, and executing the commands entered by the user.

In modern general-purpose operating systems (including Linux, Windows, and macOS), the CLI (specifically the shell) is **strictly a user-space program** and is almost never embedded directly into the operating system kernel. While the kernel provides the underlying system call interface that the shell uses to request privileged operations, the shell itself runs in user mode. This

separation ensures system stability; if a shell crashes, the underlying kernel and hardware remain unaffected. The main goal of the CLI is to facilitate the efficient execution of commands, system management, and program control.

Operating systems typically contain multiple command interpreters from which the user can choose. These command interpreters are commonly known as **shells**.

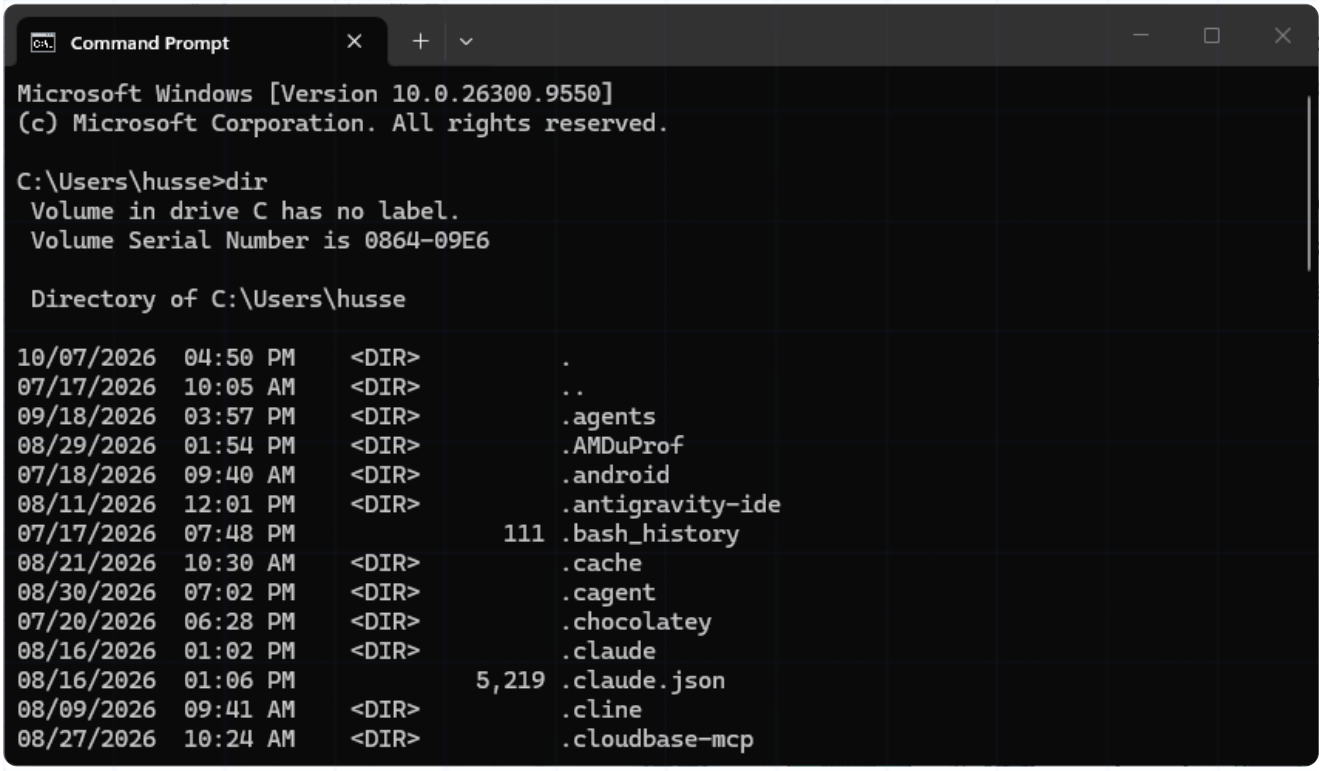


Figure 4: The Windows Command Prompt, also known as cmd.exe, is the native text-based command-line interface (CLI) for Microsoft Windows. It provides users with direct, system-level access to execute utilities, manage files, troubleshoot network issues, and automate tasks through batch scripts

3.1 Distinguishing Terminals and Shells Across Operating Systems

To understand how the CLI works in practice, it is crucial to distinguish between the **Terminal Emulator** (the graphical window) and the **Shell** (the program interpreting the text).

Here is how these components manifest across different operating systems:

1. Linux (The Primary Focus for this Lab)

- **Terminal Emulators (The Window):** GNOME Terminal, Konsole, xterm, Alacritty, Terminator.
- **Shells (The Interpreter):Bash** (Bourne Again Shell - the default on most distributions), **Zsh** (Z Shell), **sh** (Bourne Shell), and **fish** (Friendly Interactive Shell).

2. macOS

- **Terminal Emulators:** The native `Terminal.app`, iTerm2, Alacritty.
- **Shells:Zsh** (the default on modern macOS), **Bash** (the default on older macOS versions). *(Note: macOS is Unix-based, so its shell environment behaves very similarly to Linux).*

3. Windows

- **Terminal Emulators:** Windows Terminal (the modern unified app), the legacy Command Prompt window, or the PowerShell window.
- **Shells:CMD.exe** (Command Prompt - legacy MS-DOS syntax), **PowerShell** (the modern, object-oriented default), and **Git Bash** (a popular third-party Unix-like shell environment for Windows).

The Execution Pipeline: Regardless of the operating system, the relationship between the user, the terminal, and the kernel follows a structured pipeline:

$$\text{User Input} \longrightarrow \underbrace{\text{Terminal Emulator}}_{\text{Graphical Interface}} \longrightarrow \underbrace{\text{Shell (e.g., Bash / CMD)}}_{\text{Command Parsing \& Logic}} \longrightarrow \text{OS Kernel}$$

4 Computer Startup and the Linux Boot Process

When a computer starts up or reboots, a bootstrap program is executed. In modern systems, this firmware is stored in non-volatile memory such as **Flash memory or UEFI NVRAM** (replacing older ROM or EPROM technologies).

The function of this firmware (commonly known as **BIOS or UEFI**) is to perform a Power-On Self-Test (POST) to initialize and verify hardware components, such as system RAM and storage devices. Once the hardware is ready, the firmware locates and loads a **bootloader** from the storage drive. In Linux systems, this is typically **GRUB2** (GRand Unified Bootloader).

The bootloader's primary job is to load the **Linux Kernel** into memory and pass control to it. The kernel then initializes core hardware drivers and mounts the root file system. Finally, the kernel starts the first user-space process, known as the **init system** (modern Linux distributions predominantly use **systemd**). This init system configures background services, sets up the environment, and ultimately presents the user with a login prompt or graphical interface, making the computer ready to execute user applications.

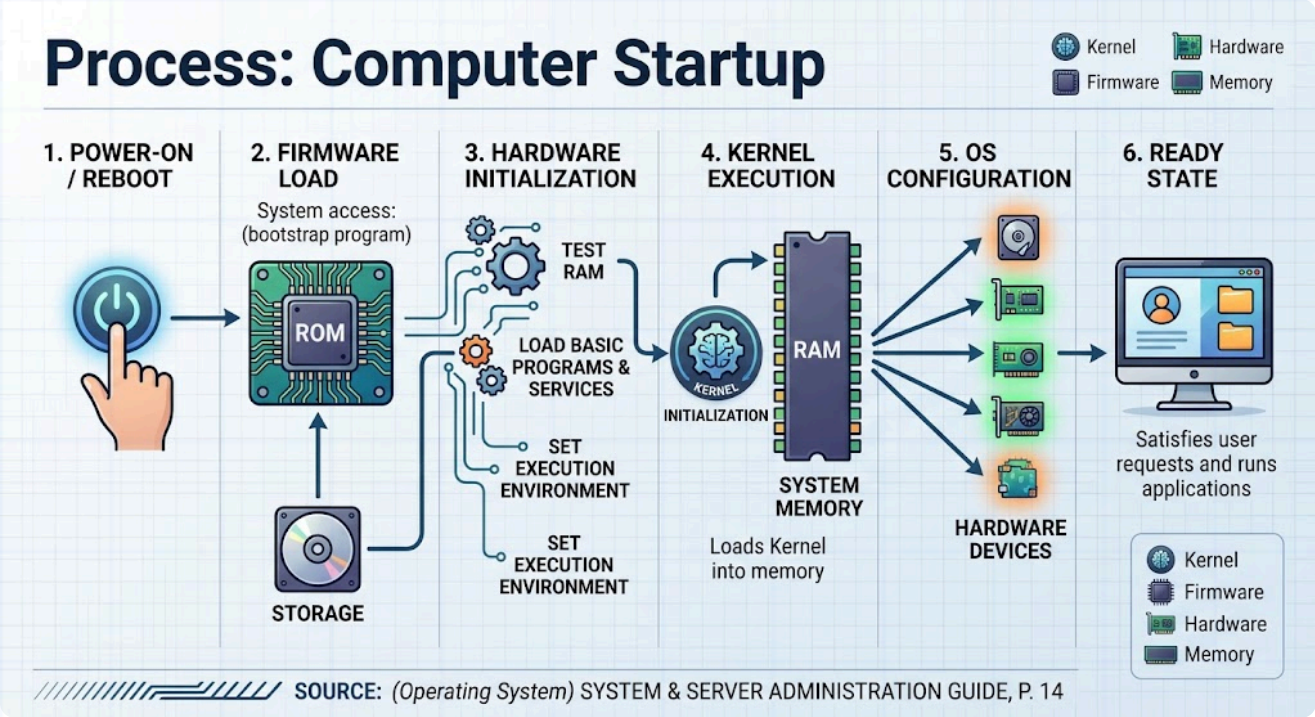


Figure 5: Computer Startup

2 The Linux System: Core Concepts and Components

1 What is Linux?

Linux is an open-source, Unix-like operating system kernel created by **Linus Torvalds** in 1991. Paired with open-source utilities and software (often referred to as GNU/Linux), it forms a complete operating system capable of managing system hardware, memory, processes, and execution environments.

Unlike proprietary operating systems such as Microsoft Windows or macOS, Linux is developed under the **GNU General Public License (GPL)**. This licensing allows anyone to inspect, modify, and redistribute its source code freely.

Linux is typically distributed as part of a **Linux Distribution (Distro)**, which packages the Linux kernel together with system utilities, package managers, user interfaces, and default applications.

Popular Distributions:

- **Enterprise & Server:** Red Hat Enterprise Linux (RHEL), Debian, Ubuntu Server, Rocky Linux / AlmaLinux.
- **Desktop & General Use:** Ubuntu, Linux Mint, Fedora, Arch Linux, Manjaro.
- **Specialized & Security:** Kali Linux (Penetration Testing), Alpine Linux (Minimal Containers).



Figure 6: Linux distributions

1.1 Global Usage & Key Statistics

While Linux holds a modest share of the traditional desktop consumer market, it dominates server, cloud, and embedded infrastructure worldwide:

- **Supercomputing: 100%** of the world's top 500 fastest supercomputers run on Linux.
- **Global Desktop Share:** Linux holds roughly **3.09% to 4.54%** of the global desktop operating system market, according to recent 2026 data from sources like Statcounter Global Stats and Wikipedia.
- **Public Cloud & Web Servers:** Over **90%** of public cloud workloads (AWS, Google Cloud, Microsoft Azure) and top web servers operate on Linux distributions.
- **Mobile Ecosystem: Android**, the world's most widely used mobile operating system, is built on top of a modified Linux kernel.
- **Containers & DevOps:** Nearly **100%** of containerized infrastructure (Docker, Kubernetes) relies on Linux kernel primitives (e.g., `cgroups` and namespaces).
- **Top Distributions:** Ubuntu leads general desktop and developer adoption at roughly **33.9%** of distributions, followed by Debian and Linux Mint.

1.2 Linux Distributions

Linux distributions (often called **distros**) are complete operating systems built on top of the **Linux Kernel**, bundled with user applications, software repositories, and system libraries.

Because Linux is **open-source**, anyone can run, study, modify, and redistribute its source code. This sets Linux apart from proprietary operating systems like UNIX, Microsoft Windows, and macOS, which are closed-source and far less customizable.

Major Linux Distribution Families:

1. **Debian Family:** Known for stability and software quality.
    - *Derived Distros*:**Ubuntu**,**Knoppix**,**Chrome OS**,**Mepis**,**Linux Mint**,**Damn Small Linux**.
2. **Red Hat Family:** Focused on enterprise and workstation environments.
    - *Derived Distros*:**Fedora**,**CentOS**,**Mandriva**,**Mageia**,**PCLinuxOS**.
3. **Slackware Family:** One of the oldest active Linux distributions, emphasizing simplicity and minimalism.
    - *Derived Distros*:**SUSE**.
4. **Other Independent Distributions:**
    - **Gentoo**,**Arch Linux**,**Puppy Linux**.

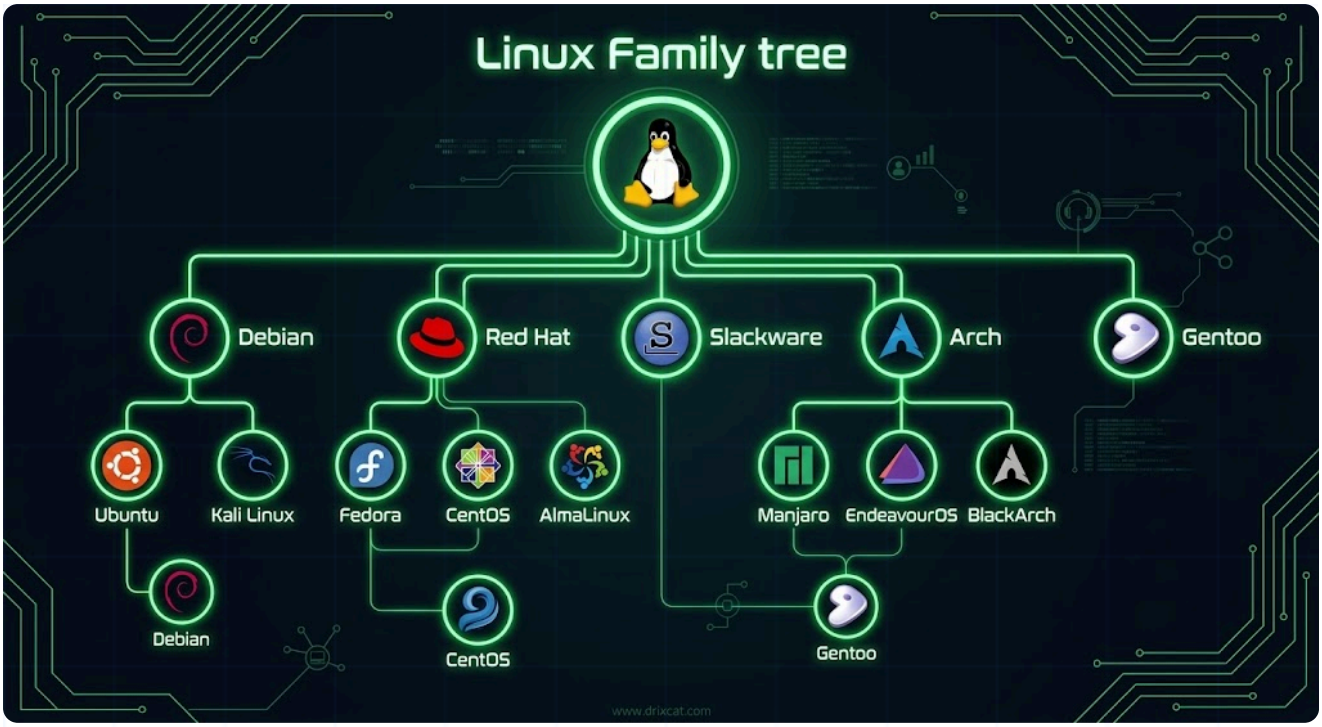


Figure 7: Linux Family Tree

1.3 Primary Advantages of Linux

- **Open-Source & Cost-Effective:** Free to use, deploy, and customize without licensing fees.
- **High Stability & Reliability:** Capable of running for years without needing a reboot, making it ideal for critical production infrastructure.
- **Granular Security:** Enforces strict multi-user access controls, privilege isolation (User Space vs. Kernel Space / Ring 3 vs. Ring 0), and mandatory access frameworks (SELinux, AppArmor).
- **Flexibility & Customization:** Highly modular—can be tailored from a minimal 5 MB headless container environment to a full desktop workstation.
- **Resource Efficiency:** Runs efficiently on legacy or resource-constrained hardware as well as high-performance servers.

1.4 Potential Challenges

- **Software Compatibility:** Certain proprietary desktop applications (e.g., Adobe Creative Cloud, native Microsoft Office) do not offer native Linux versions.
- **Hardware Driver Support:** While broad, some niche hardware or modern proprietary peripherals may require manual configuration or open-source driver workarounds.

- **Learning Curve:** Managing Linux effectively requires comfort with command-line interfaces (CLI), file permission models, and shell usage.

1.5 The Philosophy That "Everything Is A File"

In Linux, the philosophy that "everything is a file" is a fundamental concept. This concept means that in Linux, various system resources and components are represented as files or directories in the file system. This includes not only regular files but also devices, sockets, directories, and even processes.

- **Devices**: Hardware devices are represented as files in the /dev directory.
- **Processes**: Information about running processes can be accessed through files in the /proc directory.
- **Network communication**: Sockets are represented as files in the file system.
- **Configuration**: System configuration files are stored as text files in various directories.

2 Linux from the beginning - History and Evolution



2.1 The First Stage (1964–1984)

- In 1965, an agreement was formed among three large specialized companies to create an operating system that would serve researchers and other users:
  - **General Electric**, which acted as the funder for the project.
  - **Bell Laboratories**, which acted as a research institute.
  - **MIT**, which acted as an educational institute.
- In 1968, the first operating system named **Multics** was released. This system supported **Multitasking** and **Multiuser** capabilities, providing a **Time Sharing Operating System**. Despite the success of the project, ambitions were larger than the available capabilities, leading both General Electric and MIT to withdraw.

- Bell continued developing Multics under the leadership of developer Ken Thompson until 1970, when a new operating system named **Unics** was released, written in Assembly Language. The development of Unics continued until 1971, when its name was changed to **Unix**, marking a major advancement in the computer world.
- During the period 1971–1984, Unix underwent further development, was rewritten in the **C language**, and several versions were released, becoming an advanced system capable of meeting general requirements.
- In an attempt by Bell to distribute Unix, it shared it with universities, including **UC Berkeley**, where a free release of Unix was provided. This was the university where developer Ken Thompson had studied.
- In 1975, engineer Ken Thompson returned to UC Berkeley as a professor, bringing with him the latest version of Unix. Joined by two recent graduates, they worked together on tools like the **EX** text editor (which later evolved into the famous **VI** editor) and created custom applications, releasing a software distribution named **BSD** (*Berkeley Software Distribution*) in 1976.
- This prompted Bell Labs to commercialize Unix with a paid license in 1983 to protect its rights. This move angered many programmers and enthusiasts who had contributed to Unix's development over the years, leading several to resign, most notably **Richard Stallman**.

2.2 The Second Stage (1984–1991)

- After resigning, engineer Richard Stallman started developing a project called **GNU** (*GNU's Not UNIX*), aiming to build a free, open-source operating system.
- Thousands of programmers joined the project, writing applications, libraries, shells, and compilers, but they lacked a kernel and relied on the Unix kernel from 1984 through 1991.
- In 1991, a student named **Linus Torvalds** published his graduation project—a kernel compatible with AT-type hard drives that supported multithreading. This research was used to combine the software to form **GNU/Linux**.

> **Note:** The word *Linux* is derived from *Linus* (the kernel developer) and *Tux* (the penguin mascot from the developer's home city).

2.3 The Third Stage (1991–Present)

- In 1993, two major companies emerged—**Slackware** and **Debian**—which took the applications, libraries, and kernel to build their own distribution operating systems.
    - For example, **Debian** developed distributions such as **Ubuntu**, **Chrome OS**, and **Knoppix**.
- In 1994, **Red Hat** appeared, releasing **RHL** in 1998 up to version 9 in 2003, after which it was discontinued.
- In 2003, Red Hat split Linux into two branches:
    - **Fedora**: Offered for free with updates for each release lasting about 13 months.
    - **Red Hat Enterprise Linux (RHEL)**: Aimed at enterprise customers willing to pay for official support.
- Although RHEL builds are not available for free download, GNU/Linux license terms require making the source code available. This allowed the community to recompile the RHEL source code to create free enterprise-grade distributions such as **CentOS** and **AlmaLinux**.

3 Components of Linux System

The Linux OS consists of the following components:

- **Applications / GUI / Utilities / CLI:** These are the programs you interact with, ranging from graphical applications to command-line tools.
- **Shell & System Libraries:** The shell acts as your command interpreter, while libraries provide essential shared resources that applications need to function.
- **System Calls & Services:** This layer provides the interface for applications to communicate with the kernel to request resources or services.
- **Linux Kernel:** The core of the operating system, which handles vital tasks like file management, process scheduling, controlling hardware via drivers, and managing network communication.

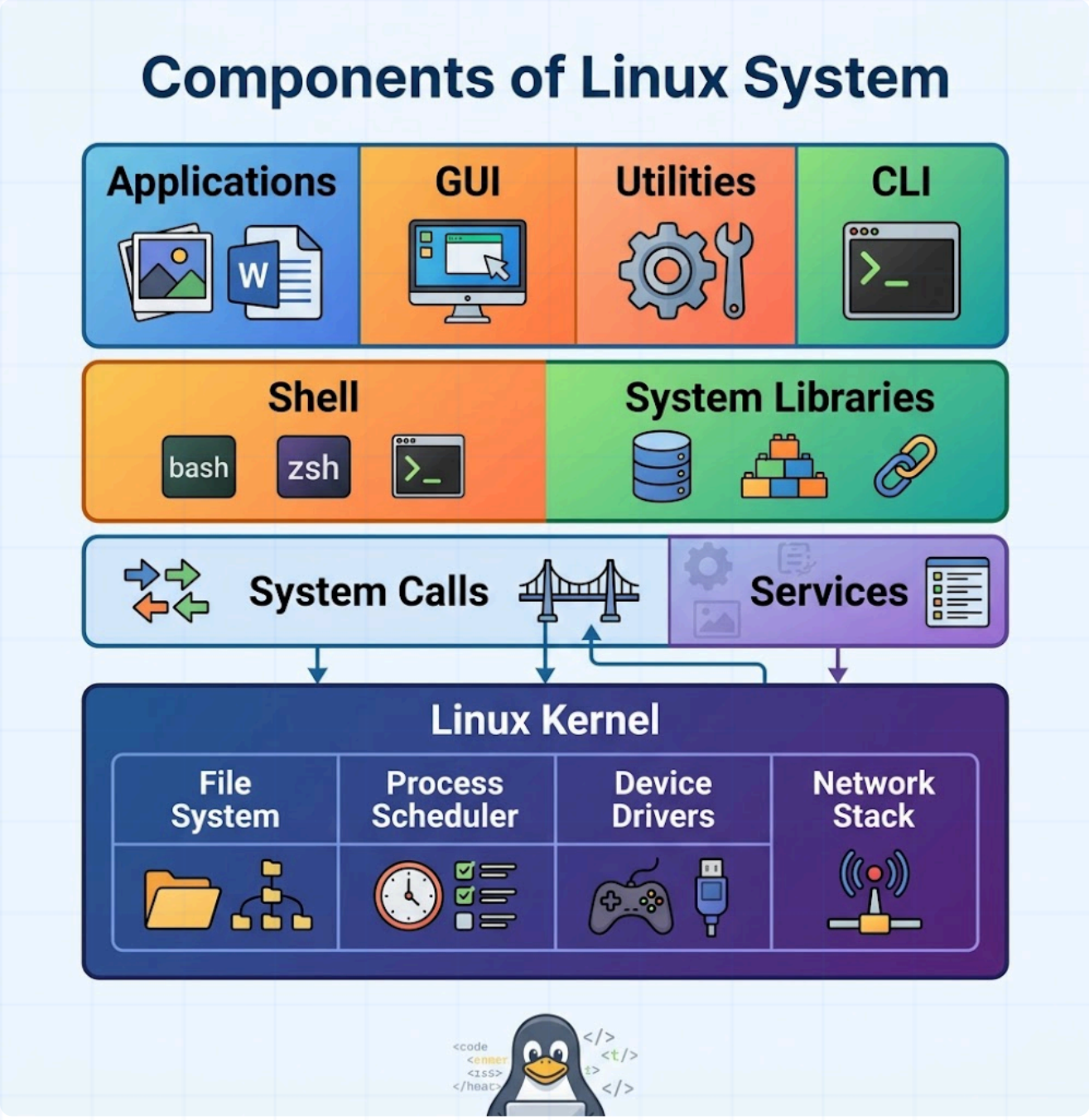


Figure 8: The basic components of a Linux system

3.1 Linux Kernel

The Linux Kernel is the core part of the operating system that establishes communication between hardware and software. For example, the kernel manages memory allocation to run specific programs and manages system resources in general.

It has four main responsibilities:

1. **Device Management:** The system contains many connected hardware devices, such as CPU, memory units, sound cards, graphics cards, etc. The Linux Kernel stores information about all

connected devices in Device Drivers. Without the Linux Kernel, it would not be possible to control these devices. The kernel knows how to interact with devices and process their requests to achieve optimal performance, manages communication between all devices, and enforces rules that all devices must follow.

2. **Memory Management:** The kernel tracks used and unused memory and ensures that processes do not interfere with each other's data by using Virtual Memory Addresses.

3. **Processes Management:** A running program is referred to as a process. The kernel acts as an intermediate layer that manages the efficient and fair distribution of system resources (CPU, memory, devices, storage) among multiple concurrent processes. It abstracts physical system resources into dedicated virtual resources for each process, making each process appear as if it is the only one running on the machine, while preventing and mitigating conflicts between processes.

4. **Handling System Calls:** Allows developers to write queries or requests for the kernel to execute specific tasks.

The Linux Kernel consists of the following Core Subsystems:

1. **Process Scheduler:** Controls the fair distribution of CPU time among all processes running concurrently on the system and determines which process gets priority to run.

2. **Memory Management Unit (MMU):** Manages the distribution and allocation of memory resources among different running processes, providing independent virtual address spaces for each process so they do not interfere with one another.

3. **Virtual File System (VFS):** Provides a unified interface to access data stored across different file systems and physical storage media, allowing users and programs to perform operations such as read, write, delete, and modify.

3.2 System Libraries

System libraries act as subprograms that assist in accessing and utilizing kernel features. They provide shared functions and software components to interact with the system and its kernel services. An example is the `Glibc` (GNU C Library), which provides APIs to interact with the system.

3.3 Shell

The Shell acts as a translator between the user and the kernel, providing an interface to input and execute commands on the system. It allows invoking programs, managing storage, and processing data. The most popular shell in Linux is the **Bash Shell** (*Bourne Again Shell*).

The shell lets us interact with Linux via a **Terminal** or **TTY**, allowing execution of binary files to create running processes and interaction with the Linux file system.

3.3.1 Scripting & Commands

- **Scripting:** Allows executing multiple tasks sequentially using files ending with the `.sh` extension.
- **Built-in Commands:** Internal commands integrated directly inside the shell (e.g., `pwd`, `help`, `type`, `set`, `unset`).
- **External Commands (Command Line Utilities):** Programs with their own binary files located in the file system (typically in `/bin`, `/sbin`, `/usr/bin`, `/usr/sbin`) independent of the shell.

3.4 System Concepts & Interaction Interfaces

To effectively navigate and manage a Linux environment, it is essential to understand how the system is deployed, how users interact with command interfaces, and how multiple terminal sessions operate.

3.4.1 Minimal vs. Maximal Deployments

Linux installations typically fall into two main environment configurations depending on their intended use case:

- **Minimal System:** A lightweight installation that operates purely through a Command Line Interface (CLI) without a graphical desktop environment. This setup is standard for servers and cloud instances to minimize resource overhead and maximize performance.
- **Maximal System:** A full-featured installation that includes both a Graphical User Interface (GUI) and a Command Line Interface (CLI). This environment is typically used on personal desktop computers and workstations.

3.4.2 Terminal vs. TTY vs. PTS

While these terms are often used interchangeably in everyday conversation, they refer to distinct technical interface types:

- **Terminal:** In a **Maximal (GUI)** system, a terminal refers to the graphical terminal emulator window (such as GNOME Terminal or Konsole) used to run a shell.
- **TTY (Teletypewriter):** In a **Minimal (non-GUI)** system, the native text interface is called a TTY. Historically derived from physical electromechanical teleprinters, a TTY provides direct physical text access to the machine.
- **PTS (Pseudo Terminal Slave):** A virtual terminal session created when accessing a Linux machine remotely (e.g., via SSH) or inside a graphical terminal emulator.

> ⓘ **Note**
>
> **TTY act as a port (or a communication stream)** that sends and receives data to and from the operating system. In Linux and Unix-like operating systems, a TTY acts as a bidirectional communication channel between a user interface (or program) and the kernel.

How TTY Functions as a Data Port

1. "Everything is a File" (Device Files)

In Linux, every TTY interface is represented in the file system as a character device file located under `/dev/`:

- **Physical Console:** `/dev/tty1`, `/dev/tty2`, etc.
- **Pseudo-Terminals (SSH / Terminal Windows):** `/dev/pts/0`, `/dev/pts/1`, etc.

Because it is represented as a device file, the operating system interacts with it using standard input/output stream operations (`read` and `write`):

- **Writing Data (Transmitting):** When a process writes data to `/dev/pts/0`, the operating system routes that data through the TTY driver to be rendered on the screen.
- **Reading Data (Receiving):** When keypresses or data are sent into `/dev/pts/0`, the operating system reads those characters and passes them as `stdin` (standard input) to the running shell or application.

📝 EXAMPLE 1: Open Ubuntu Terminal in WSL2 & Identify

Launch **Windows Terminal** from your Start menu and select **Ubuntu** (or type `wsl` in Windows PowerShell).

- Run the `tty` command: `tty`
- **Observation:** The output displays `/dev/pts/2`.
  - **Why?**

✅ Show Solution

Terminal acts as a graphical client attached to a virtual pseudo-terminal slave (`pts`) managed inside WSL2.

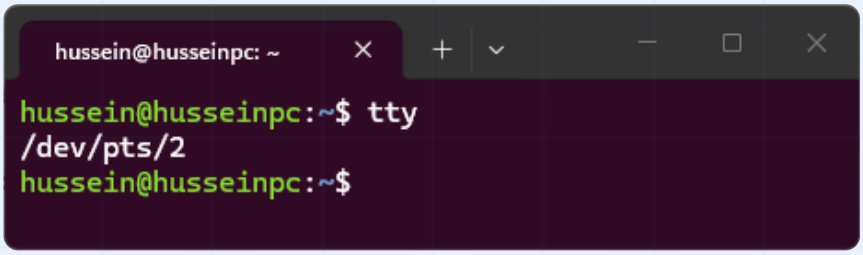


Figure 9: Open Ubuntu Terminal in WSL2 & Identify

📝 EXAMPLE 2: Open a New Tab and Send Data Between PTS Sessions

Because WSL2 runs inside a virtualized environment, each new terminal tab or pane creates a distinct **PTS device**.

1. **Open a Second Tab:** Press `Ctrl + Shift + T` (or click the `+` button in Windows Terminal) to open a second Ubuntu tab.
2. In **Tab 2**, run `tty` to check its unique device name.
3. **Send Data from Tab 2 to Tab 1:** In **Tab 2**, write text directly to Tab 1's device file (`/dev/pts/0`):

```bash
echo "Hello from Tab 2!" > /dev/pts/0
```

Show Solution

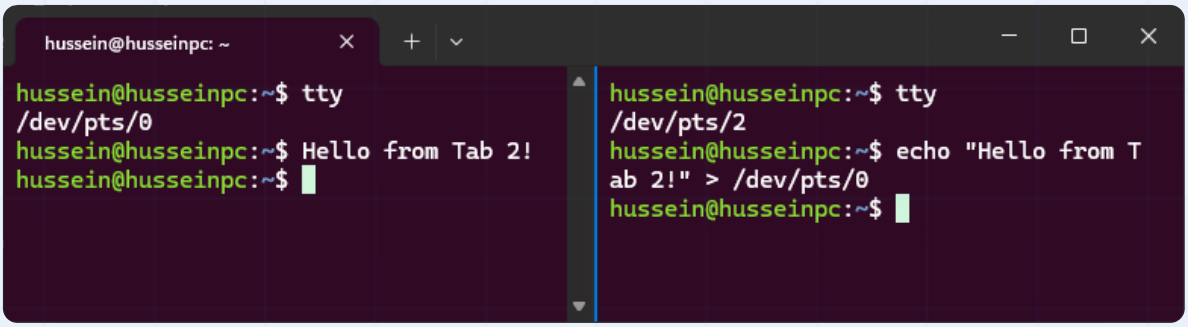


Figure 10: Open a New Tab and Send Data Between PTS Sessions

3.4.3 Virtual TTYs

Linux is a multi-session operating system that supports multiple concurrent local login sessions. These are organized into **Virtual TTYs**:

- **Switching Sessions:** You can switch between different virtual TTYs using the keyboard shortcut `Alt + Ctrl + Fn` (where `n` represents the TTY number, typically ranging from `1` to `6` or `9` depending on the distribution). Alternatively, you can switch sessions using the command `chvt n`.
- **Identifying Your Active TTY:** To check which virtual terminal session you are currently using, execute the `tty` command.
- **Ending a Session:** To log out of the current virtual TTY session, execute the `exit` command.

4 Basic Linux Commands

Before diving into the commands, it is essential to understand the general syntax for Linux commands:

```bash
command [option] [arguments]
```

- `command` : The action or program you want to execute.

- `option` *(optional)*: Used to modify the behavior of the command.
- `arguments` *(optional)*: The targets (files, directories, processes, etc.) that the command applies to.

📝 EXAMPLE 3: clear Command

clear: Clears all previous output from the terminal screen.

```bash
clear
```

📝 EXAMPLE 4: jobs Command

Displays all active background jobs along with their statuses. A job is essentially a process managed directly by the shell.

```bash
jobs
```

📝 EXAMPLE 5: top Command

Equivalent to the Task Manager in Windows. It displays real-time running processes and their CPU and memory usage. It is very useful for monitoring system resources and identifying resource-heavy processes to terminate.

```bash
top
```

✅ Show Result

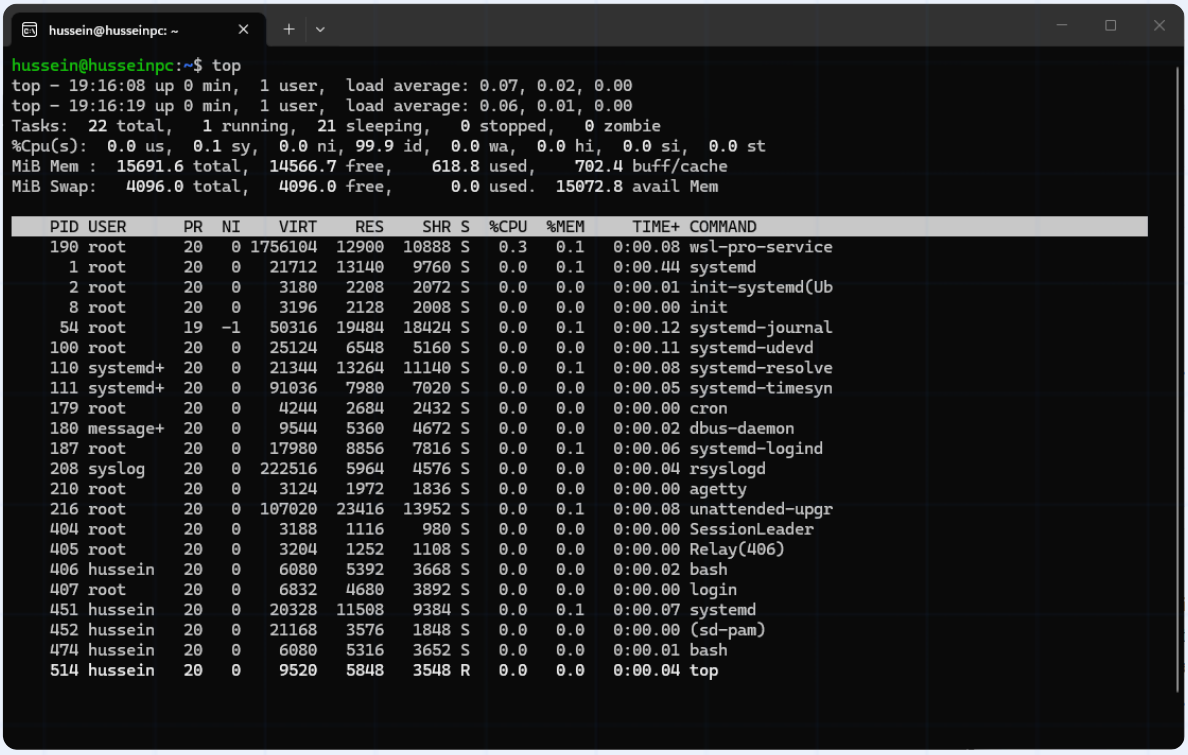


Figure 11: Top Command

EXAMPLE 6: history Command

Lists all previously executed commands in the current shell session, making it easy to review or rerun past commands.

```bash
history
```

Show Result

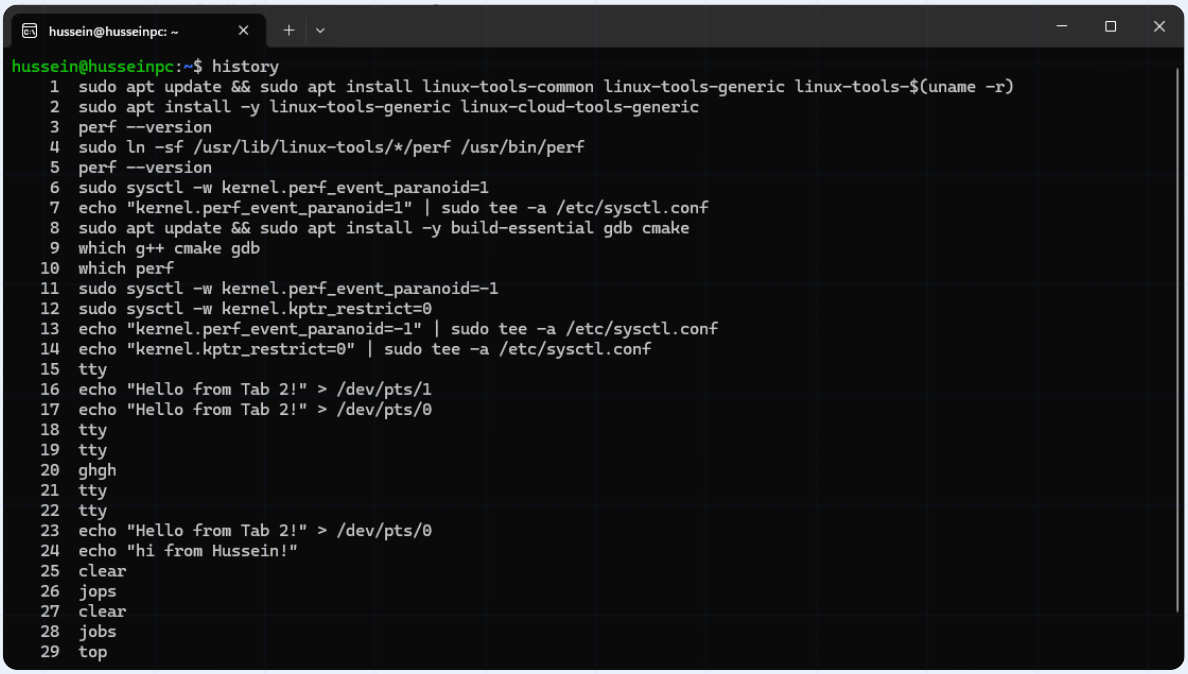


Figure 12: History Command

EXAMPLE 7: man Command

Displays the manual/documentation page for a specific command to explain its usage, options, and parameters.

```bash
man clear
```

EXAMPLE 8: id Command

Displays user identity information, including the User ID (`UID`), Primary Group ID (`GID`), and all Secondary Groups the user belongs to.

```bash
id
```

✅ Show Result

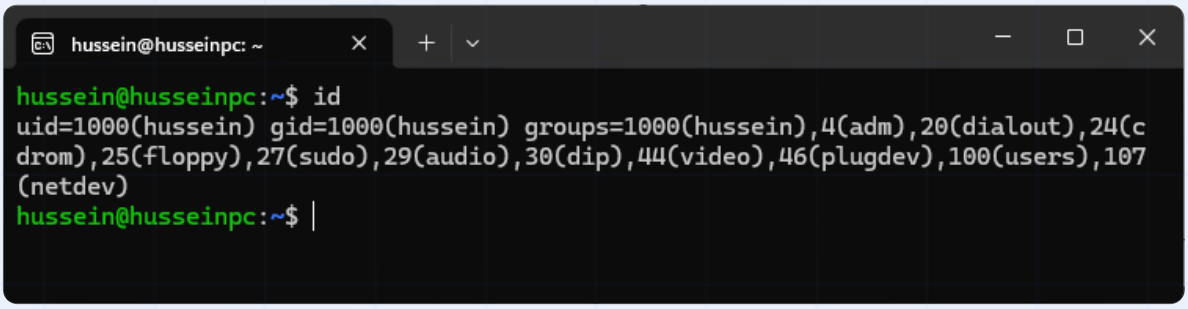


Figure 13: id command

EXAMPLE 9: echo Command

Prints a line of text or variable output directly to the terminal screen.

```bash
echo "Hello, my name is Hussein"
```

Show Result

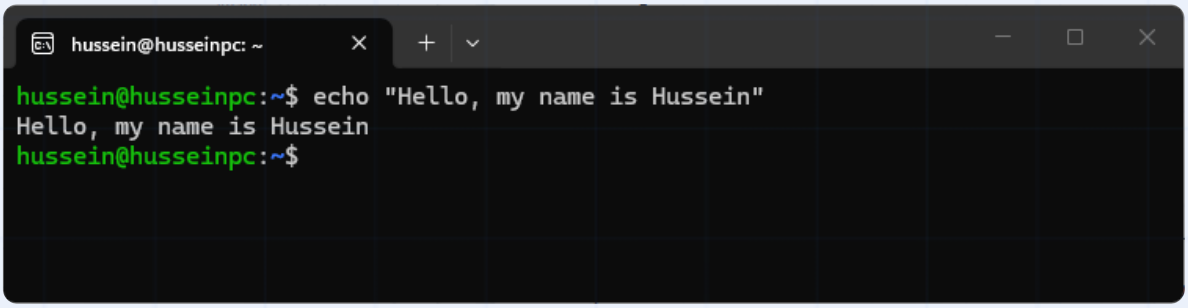


Figure 14: echo command

EXAMPLE 10: hostname Command

Displays the network host name of the system. Adding the `-i` flag displays the system's IP address.

```bash
```

```
hostname
hostname -i
```

Show Result

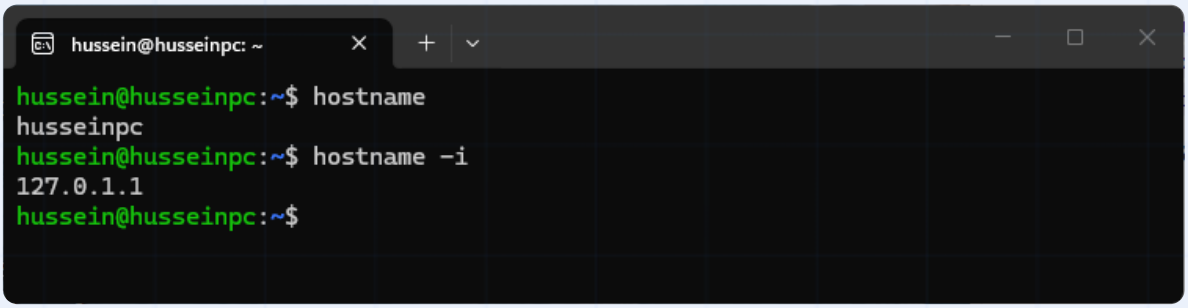


Figure 15: hostname command

5 Use Linux on a Windows Machine

As a developer, you might need to run both Linux and Windows side by side. Luckily, there are several ways you can get the best of both worlds without getting different computers for each operating system.

5.1 Option 1: "Dual-boot" Linux + Windows

With dual boot, you can install Linux alongside Windows on your computer, allowing you to choose which operating system to use at startup. This requires partitioning your hard drive and installing Linux on a separate partition. With this approach, you can only use one operating system at a time.

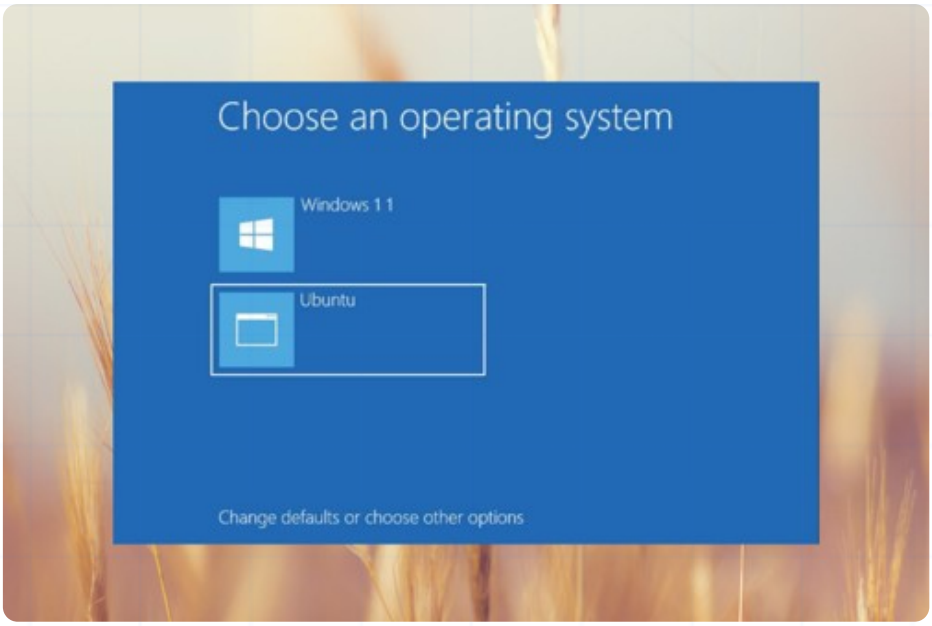


Figure 16: "Dual-boot" Linux + Windows

5.2 Option 2: Use Windows Subsystem for Linux (WSL2)

- Windows Subsystem for Linux provides a compatibility layer that lets you run Linux binary executables natively on Windows.
- The setup for WSL is simple and not time-consuming.
- It is lightweight compared to VMs where you have to allocate resources from the host machine.
- You don't need to install any ISO or virtual disc image for Linux machines which tend to be heavy files.
- You can use Windows and Linux side by side.

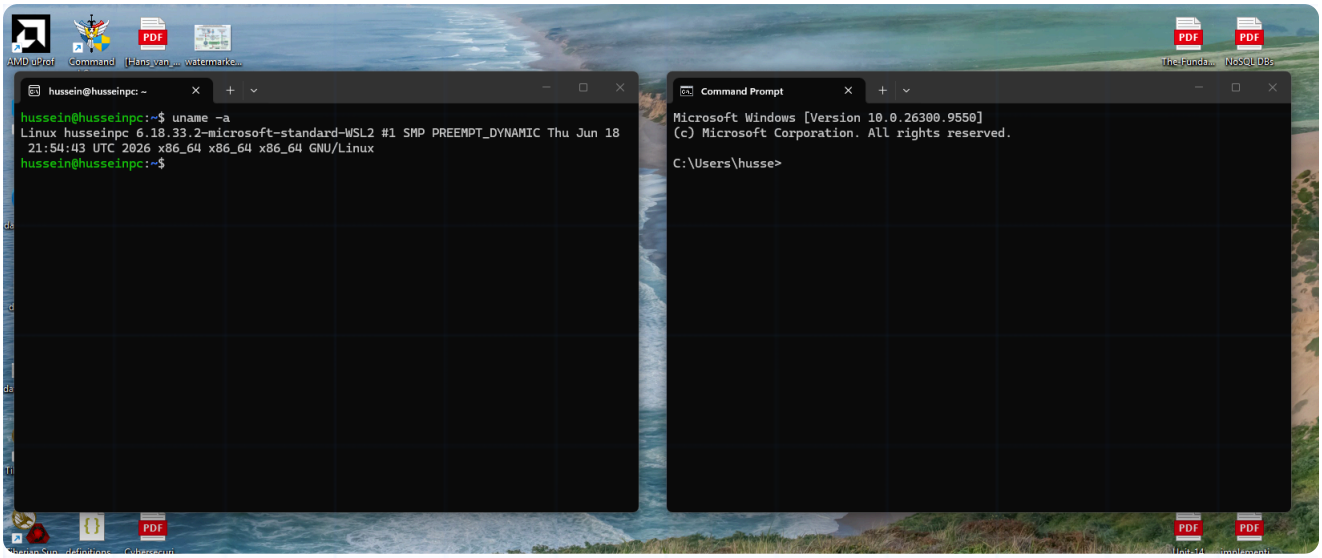


Figure 17: Use Windows Subsystem for Linux (WSL2)



5.3 Option 3: Use a Virtual Machine (VM)

- A virtual machine (VM) is a software emulation of a physical computer system. It allows you to run multiple operating systems and applications on a single physical machine simultaneously.
- VM software provides options to allocate and manage hardware resources for each VM, including CPU cores, memory, disk space, and network bandwidth. You can adjust these allocations based on the requirements of the guest operating systems and applications.

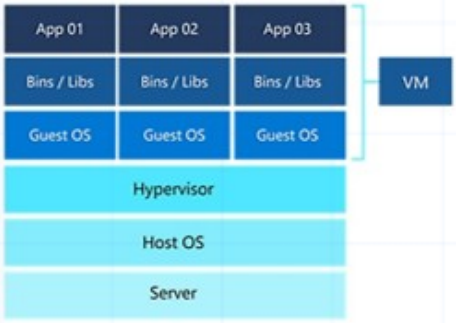


Figure 18: Virtual Machine (VM)

Free Virtual Machine Software

You can use Oracle VirtualBox, which is a popular, free, and open-source platform package released under the GPL version 3 license. It allows you to create and run multiple virtual machines (guest computers) on your host operating system. You can download it directly from the Downloads – Oracle VirtualBox page by choosing your host OS (Windows, macOS, or Linux distributions).

5.3.1 What is Virtualization

📖 DEFINITION 1: **Virtualization**

Virtualization is the process of creating a virtual (rather than actual) version of something, such as hardware platforms, storage devices, or network resources.

In computing, virtualization uses software—known as a **hypervisor**—to abstract physical hardware. This allows multiple virtual machines (**Guest Computers**) to run independently on a single physical machine (**Host Computer**), sharing its CPU, memory, and storage resources.

A hypervisor is a software layer that allows multiple operating systems to run on a single physical hardware platform. It creates and manages virtual machines (VMs), which are isolated environments that can run
their operating systems independently of each other. The hypervisor abstracts the physical hardware, allocating resources to each VM and ensuring they operate efficiently and securely. There are two main types of hypervisors:

- Type 1 (bare-metal) hypervisors run directly on the host hardware
- Type 2 (hosted) hypervisors run on a conventional operating system like Windows or Linux.

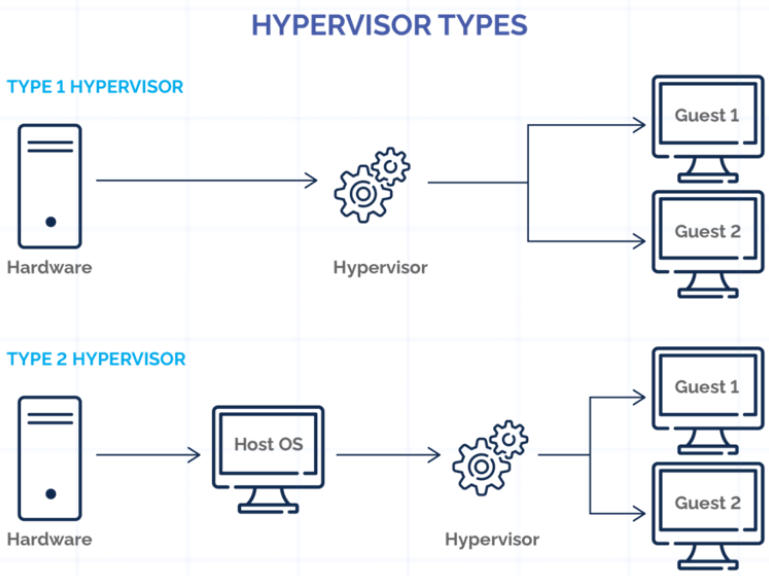


Figure 19: hypervisor Types

3 File Systems and Their Commands

1 Introduction to File Systems

In the Linux operating system, "everything is either a file or a process." This philosophy goes beyond traditional text files, images, and programs—it also includes disk partitions, hardware device drivers, and directories.

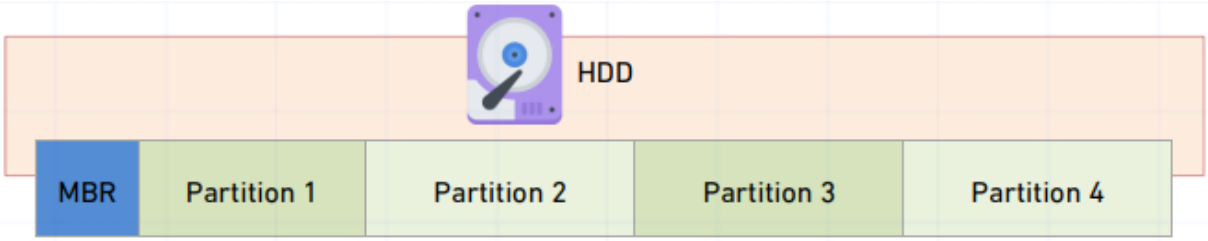
> 📖 DEFINITION 1: **File System**
>
> A **file system** is the component responsible for controlling how data is stored, organized, and retrieved on a storage device.
>
> Without a file system, all the information stored on a hard drive or SSD would just be an indistinguishable, single block of raw data, making it impossible to tell where individual files begin and end.

1.1 Key Functions of a File System

- **Data Organization:** It structures data into logical units called **files** and **directories (folders)**, arranging them in a hierarchical tree structure.
- **Metadata Management:** It keeps track of essential attributes for every file, such as its name, size, creation/modification timestamps, access permissions, and ownership.
- **Storage Allocation:** It maps out where files are physically located across the storage blocks of a disk partition, ensuring space is allocated efficiently.

1.1.1 Disk Partitioning

> 📖 DEFINITION 2: **Disk Partitioning**
>
> A raw storage device initially appears as empty, raw space. Through partitioning, it is logically divided into independent segments called **Partitions**.



Partitioning can be performed using standards such as:

- **MBR (Master Boot Record):** Allocates a small space at the beginning of the drive to hold startup metadata and the partition table.*(Crucial Note: If the partition table is lost, all data is lost because it maps the start and end points of each partition.)*
- **GPT (GUID Partition Table).**

Every partition is managed individually by a **File System** (such as FAT32, NTFS, ext2, ext3, ext4, or XFS). Without a file system, stored information would be an unidentifiable, single piece of data.

2 Files in Linux

In the Linux operating system, **"everything is a file; if it is not a file, it is a process."** The term "file" goes beyond traditional text documents, images, and programs—it also encompasses partitions, hardware device drivers, and directories.

2.1 File Characteristics & Attributes

The file system provides every file with a set of essential properties and information, including:

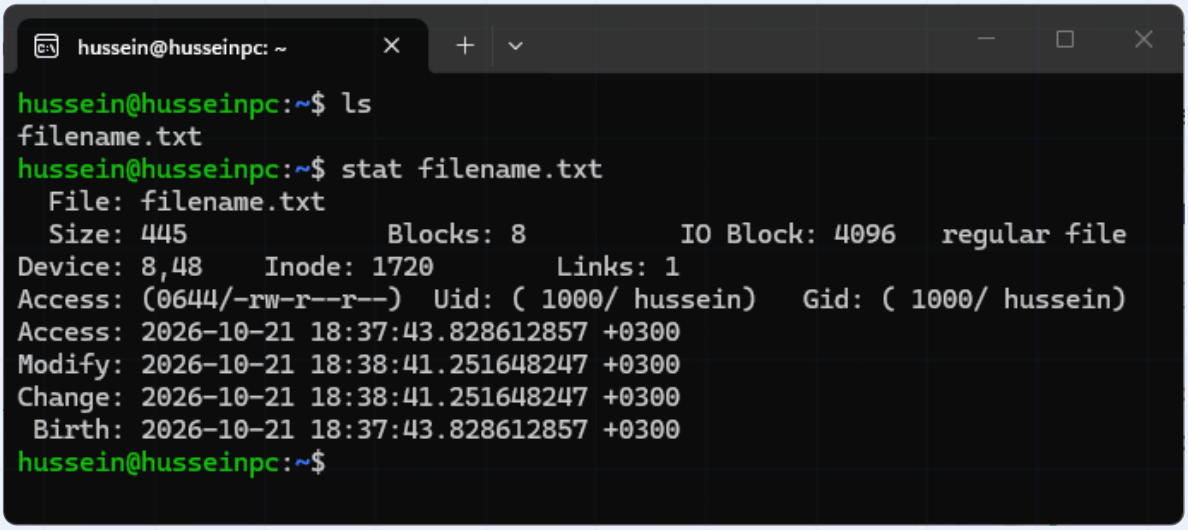
- **File-Name:** The name given to a file to identify and access it within the system. It is defined by the user or the system and must be unique within its parent directory.
- **Metadata:** Descriptive data about the file, such as its size, creation date, last modification timestamp, access permissions, owner, and associated user group. Metadata assists in managing files and determining access rules.
- **Inode (Unique Number):** A unique identifier assigned to every file in the file system. The Inode contains file metadata and points to the physical location of the file's data blocks on the disk. The system uses the Inode for internal operations instead of the file name.

> 📝 EXAMPLE 1: **Illustrating File Metadata & Inodes (stat)**

The `stat` command is the best tool to inspect a file's detailed attributes, including its **Inode number**,**Metadata** (such as size, timestamps, and access permissions), and block information.

```bash
stat filename.txt
```

✅ Show Solution



Breakdown of the Output:

1. **File Type & Size:** Indicates it is a `regular file` with a size of `445` bytes occupying `8` blocks on the disk.
2. **Inode Number (`Inode: 1720`):** The unique system index number that the Linux kernel uses internally to reference this file instead of its human-readable name.
3. **Permissions & Ownership (`Access: (0644/...`, `Uid/Gid`):** Shows that the owner `hussein` (UID 1000) has read/write permissions, while members of the group and others have read-only access.
4. **Timestamps:**
   - **Access:** The last time the file was read.
   - **Modify:** The last time the file's content was altered.
   - **Change:** The last time the file's metadata or attributes were modified.
   - **Birth:** The exact creation timestamp of the file.

2.2 File Types

File types indicate the operating system's ability to distinguish between different categories of data. Linux supports several file types:

1. Regular Files (`-`)

- Also known as **General Files**, these are standard packages of information with a specific name and extension.
- They are used to store data such as text documents, code, and images. Users can perform standard operations like adding, modifying, deleting, or removing these files entirely. They are represented by a dash (`-`).

2. Directory Files (`d`)

- Directories (folders) act as containers that hold other files and subdirectories.
- They are organized in a hierarchical, tree-like structure, meaning a directory can contain other directories within it. They are represented by the letter (`d`).

3. Special / Device Files

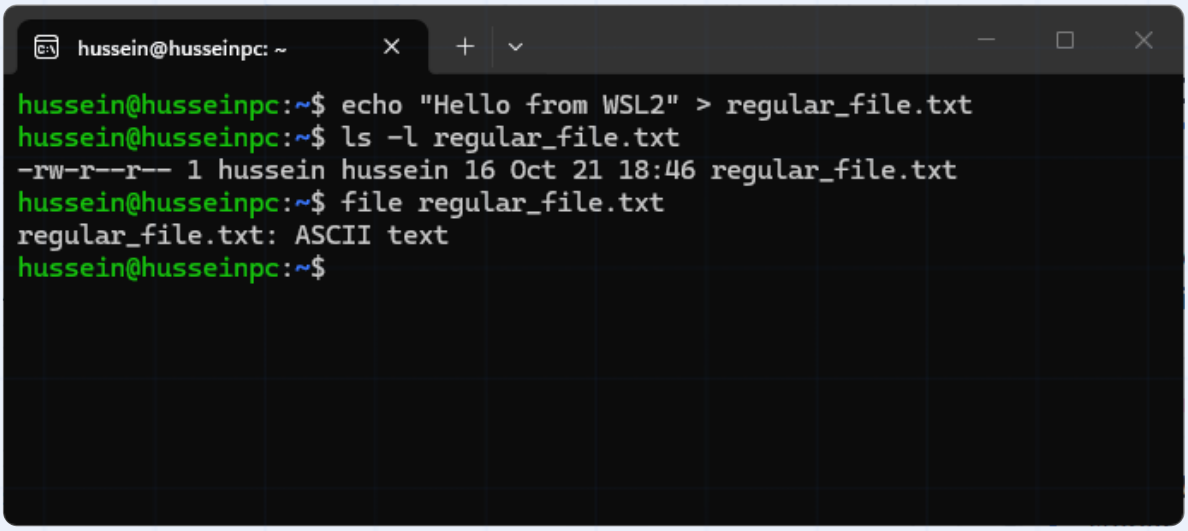
- These are files used for specialized purposes, often referred to as **Device Files**.
- Unlike operating systems like Windows—which represent hardware such as hard drives and CD-ROMs with drive letters like `C:` or `D:` —Linux represents hardware devices as special files within the file system:
  - **Block Files (`b`):** Represent block storage devices (like hard drives and SSDs) that transfer data in fixed blocks.
  - **Character Device Files (`c`):** Represent devices that transfer data as a stream of characters (like keyboards and mice).
  - **Named Pipe Files (`p`):** Used for inter-process communication (IPC).
  - **Symbolic Link Files (`l`):** Pointer files that reference another file or directory.

> 📝 EXAMPLE 2: **Regular Files (-)**
>
> Regular files are standard data files (like text documents, code, or scripts) represented by a dash (`-`) in the `ls -l` output.

```bash
# Create a regular text file
echo "Hello from WSL2" > regular_file.txt

# Inspect file type and permissions
ls -l regular_file.txt
file regular_file.txt
```

✅ Show Solution



- The leading `-` in `ls -l` indicates it is a **regular file**.
- The `file` command correctly identifies it as text data.

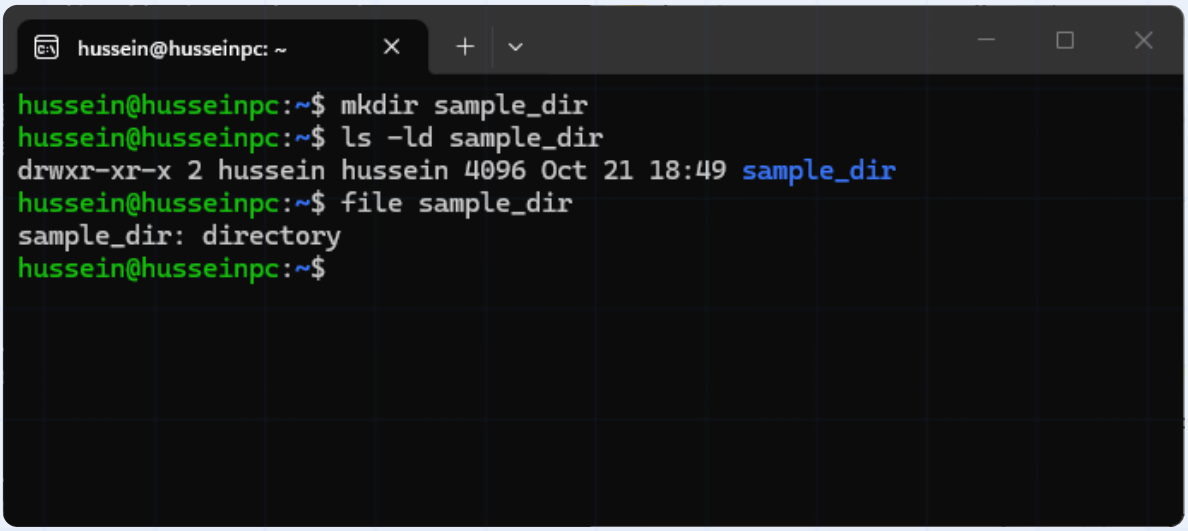
📝 EXAMPLE 3: Directory Files (d)

Directories act as containers holding other files and subdirectories, represented by the letter `d`.

```bash
# Create a directory
mkdir sample_dir

# Check its file type attributes
ls -ld sample_dir
file sample_dir
```

✅ Show Solution



The leading `d` indicates it is a **directory file**.

3 File Structure

Unlike Windows systems—which store files across separate partition drives such as `C:`, `D:`, and `E:`—Linux organizes all files and directories into a single, unified hierarchical tree structure starting from the **Root Directory (`/`)**.

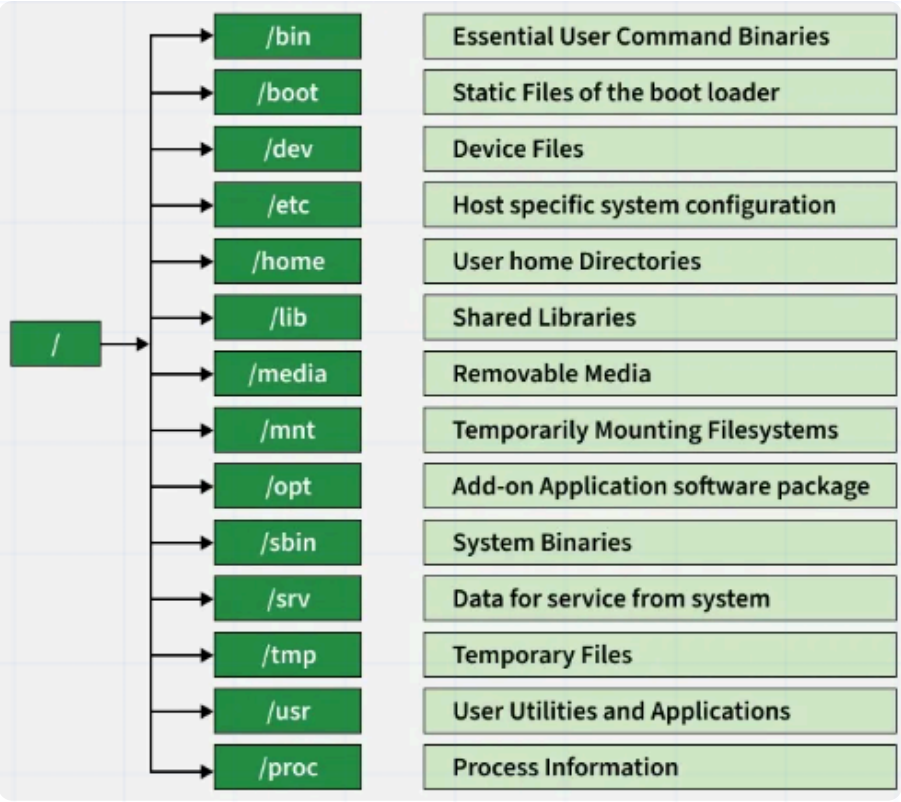


Figure 20: Linux File Hierarchy Structure

Linux relies on a **hierarchical file structure** where every file and folder branches out from a single top-level directory called the **Root Directory**.

- **Root Directory (`/`):** Represented by a forward slash (`/`), it serves as the absolute entry point to the entire file system. Everything on the system—including attached storage devices, mounted partitions, and system configurations—lives under this root directory.
- **User Home Directories (`/home`):** Every user is assigned a dedicated home directory (e.g., `/home/username`) to store personal files, desktop items, and downloads.

3.1 Standard Directory Categories (FHS)

Linux organizes system and user data into specialized standard directories based on their purpose:

- **Binary Directories (البرامج الثنائية):** Contains compiled source code and executable program files.
  - `/bin` : Essential user binaries and common Linux commands available to all users.
  - `/sbin` : System binaries and administrative commands primarily used by the root user.
  - `/lib` : Essential shared libraries required by binaries in `/bin` and `/sbin` .

- **Configuration Directories (مجلدات التكوين):** Contains system and boot configuration files.
  - `/boot` : Contains bootloader and kernel configuration files.*Crucial Note: Modifying or deleting files here can cause the system to fail to boot.*
  - `/etc` : Contains system-wide configuration files (such as network settings, user account data, and application parameters).
- **Data Directories (مجلدات البيانات):** Stores persistent user and service data.
  - `/home` : User home directories.
  - `/tmp` : Temporary storage directory; data here is volatile and can be cleared after system reboots.
  - `/srv` : Contains data for services provided by the system (such as web or FTP data).
- **Memory & Virtual Directories (مجلدات الذاكرة):** Dynamic pseudo-filesystems that do not take up actual hard disk space.
  - `/dev` : Device files used to represent hardware components (e.g., `/dev/sda` for hard drives).
  - `/proc` : Virtual filesystem containing runtime information about active system processes and the kernel.
  - `/sys` : Provides detailed information and statistics about kernel configurations and connected hardware.
- **Variable & Shared Directories:**
  - `/var` : Variable directory containing files that constantly change in size (such as logs, cache, and spool files).
  - `/usr` (Unix System Resources): Contains shareable, read-only data, applications, fonts, and themes.
  - `/media` : Default mount point for removable storage media like USB flash drives and CD-ROMs.

3.2 Absolute vs. Relative Paths

When navigating the file system, paths can be defined in two ways:

- **Absolute Path (المسار المطلق):** A complete path starting explicitly from the root directory (`/`) and tracing all subdirectories down to the target (e.g., `/home/hussein/Documents`).
- **Relative Path (المسار النسبي):** A path relative to your current working directory. It never starts with a slash `/` but rather points to locations relative to where you currently stand in the terminal.

> **Note:** Linux file names and paths are **case-sensitive** (e.g., a folder named `Documents` is entirely different from one named `documents`).

> 🎬 SCENARIO 1: **how to navigate between directories using both Absolute Paths and Relative Paths in your Linux terminal (WSL2)**
>
> Let's assume your current working directory is inside your user's home folder: `/home/hussein` , and you want to navigate into a folder named `Documents` located inside `/home/hussein/Desktop/` .
>
> ### 1. Absolute Path (المسار المطلق)
>
> An absolute path **always starts from the Root Directory (`/`)** and traces the exact full path down to the destination, regardless of where you are currently standing in the terminal.
>
> ```bash
> cd /home/hussein/Desktop/Documents
> ```
>
> **How it works:** It explicitly tells the system:"*Start at the absolute root (`/`), go into* `home` *, then* `hussein` *, then* `Desktop` *, and finally into* `Documents` ." You can run this command from **anywhere** on your system, and it will take you straight to the target folder.
>
> ### 2. Relative Path (المسار النسبي)
>
> A relative path **starts from your current location** and navigates forward or backward relative to where you stand right now. It **never** starts with a forward slash (`/`).
>
> **A. Navigating Forward into a Subfolder:**
>
> If you are currently in `/home/hussein` and want to enter `Desktop/Documents` :

```bash
cd Desktop/Documents
```

- **How it works:** Because it doesn't start with `/`, Linux looks inside your *current* directory (`/home/hussein`), finds the `Desktop` folder, and goes straight into `Documents`.

B. Navigating Using Parent Directory Shortcuts (`..`):

Suppose you are currently deep inside `/home/hussein/Downloads`, and you want to jump over to `/home/hussein/Desktop/Documents`.

```bash
cd ../Desktop/Documents
```

How it works:

- The `..` tells the system to step back **one level up** (from `Downloads` back to `/home/hussein`).
- From there, it immediately steps forward into `Desktop/Documents`.